1 ANALOG INPUT

1.1 Tasks:

1. Unmount robot's bumper and all connections to the switch.
2. Equip the robot with distance sensor according to video and scheme (see fig. 1).

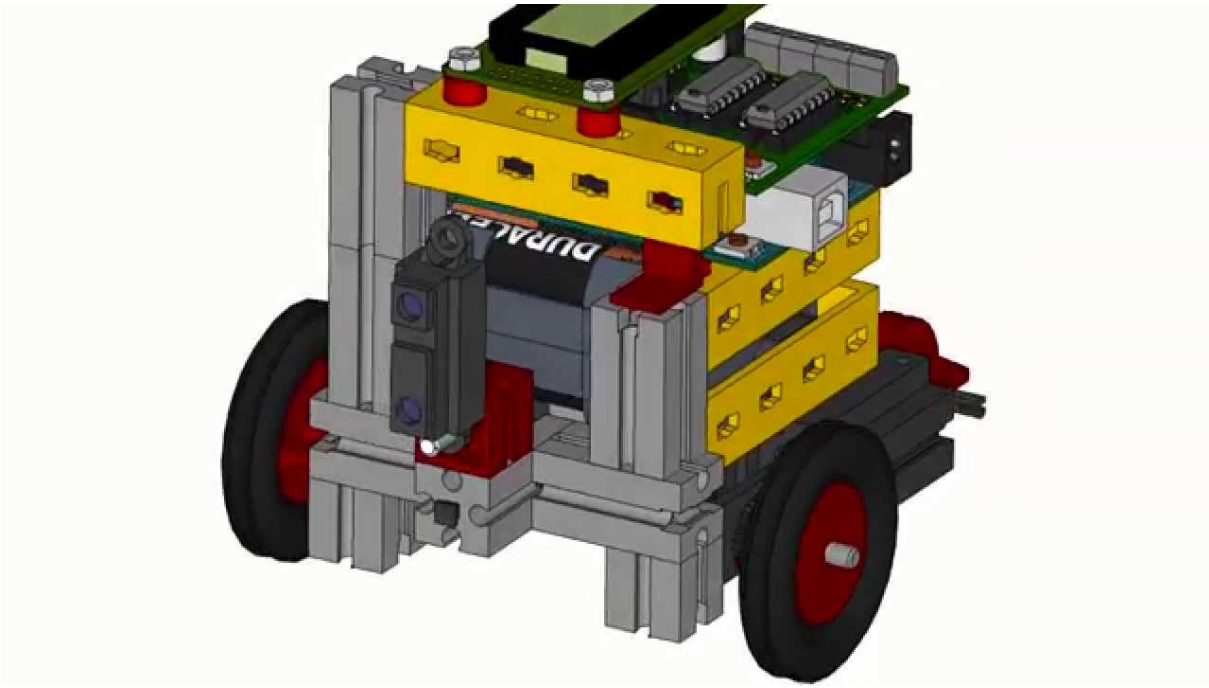

Figure 1: Mounting possition of analog distance sensor.

3. Copy & Paste next program and check the output of distance sensor in Serial monitor.

```
const int DIST_SEN_PIN = A0;
void setup()
{
  pinMode(DIST_SEN_PIN, INPUT);
 Serial.begin(9600);
}
void loop()
{
  int distance = analogRead(DIST_SEN_PIN);
 Serial.println(distance);
 delay(1000);
}
```

4. From the datasheet for the distance sensor try to code the function for measuring the distance in cm. According to documentation there is almost linear trend between output voltage and

$distance^{-1}$. Thus we can get good result with eq. 1.

$$distance^{-1}[cm] = 0.045V_{out} \tag{1}$$

Next example can be your guide to code the function.

```
float getDistance_cm()
{
  int volt_out = analogRead(DIST_SEN_PIN);
  float distance = 1/(0.045 * volt_out);
  return distance;
}
```

1.2 Questions:

1. What kind of values do you getting from the reading of the distance sensor with the function `analogRead(A0)`?
2. Find the reasonable value where you should stop the robot.
3. Measure the voltage potencial of the sensor's output.

1.3 Summary:

1.3.1 Analog to digital converter - ADC

ADC is an electronic sistem that converts analog signal (voltage) to a digitalized values. In our particular case the range of an analog voltage from 0V to 5V is converted to range of numbers from 0 to 1023.

1.4 Issues:

1.4.1 <++>

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